

Plectic action on cohomology of Hilbert modular varieties

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Introduction modular curve

$$K \subset GL_2(A_f) \rightsquigarrow Y_K / \mathbb{Q} = \{ \text{elliptic curve} + \text{level } K \}$$

$$Y_K(\mathbb{C}) = GL_2(\mathbb{Q}) \backslash (\mathbb{C} \backslash \mathbb{R}) \times GL_2(A_f) / K = \coprod \Gamma_i \backslash \mathcal{H}$$

$$H_p^1(Y_K) := \text{Im} \left(H_c^1(Y_K, \bar{\omega}, \bar{\omega}_p) \rightarrow H^1(Y_K, \bar{\omega}, \bar{\omega}_p) \right) \hookrightarrow \text{Gal} = \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$$

$$H_p^1 = \varinjlim_{K \subset GL_2(A_f)} H_p^1(Y_K) = \bigoplus_{\substack{\pi = \pi^\infty \otimes \pi_\infty \\ \uparrow \\ \text{newform of wt } 2}} \pi^\infty \otimes p_\pi \quad , \quad \dim p_\pi = 2$$

$\hookrightarrow GL_2(A_f) \times GL_1$

p_π^{ss} is determined by π .

Fact. p_π^{ss} is irreducible $\Rightarrow p_\pi = p_\pi^{ss}$.

Cor. $H_p^1(Y_K)$ is a semisimple GL_1 -representation

$$k \geq 0 \rightsquigarrow GL_2 \supset \text{Sym}^k \text{Std} \rightsquigarrow \mathbb{L}_k : p\text{-adic local system on } Y_K$$

Rem. Same holds for $H_p^1(Y_K, \mathbb{L}_k)$

$$F/\mathbb{Q} \text{ totally real}, \quad d = [F:\mathbb{Q}], \quad F \otimes_{\mathbb{Q}} \mathbb{R} \simeq \mathbb{R}^d$$

$$K \subset GL_2(A_f \otimes_{\mathbb{Q}} F) \rightsquigarrow \text{Hilbert modular variety } Sh_K / \mathbb{Q}$$

$$Sh_K(\mathbb{C}) \simeq GL_2(F) \backslash (\mathbb{C} \backslash \mathbb{R})^d \times GL_2(A_f \otimes_{\mathbb{Q}} F) / K$$

$$Sh_K : d\text{-dim'l alg. var.} / \mathbb{Q}$$

$$\text{weight } \underline{k} = (k_1, \dots, k_d; \omega) \quad k_1, \dots, k_d \geq 0, \quad k_1 \equiv \dots \equiv k_d \equiv \omega \pmod{2}$$

$$\rightsquigarrow \mathbb{L}_{\underline{k}} = p\text{-adic loc. sys. on } Sh_K$$

$$H_p^d := \varinjlim_k \operatorname{Im} \left(H_c^d(\operatorname{Sh}_k, \bar{\omega}, \mathbb{L}_k) \rightarrow H^d(\operatorname{Sh}_k, \bar{\omega}, \mathbb{L}_k) \right)$$

↳

$$H_2(A_f \otimes F) \times G_A = H_{\text{cusp}}^d \oplus H_{\text{non-cusp}}^d$$

$$H_{\text{cusp}}^d = \bigoplus_{\pi = \pi_\infty \otimes \pi_\infty} \pi_\infty \otimes \underset{\substack{\cup \\ G_A}}{V(\pi)}, \quad \dim V(\pi) = 2^d$$

$$\pi \rightsquigarrow \rho_\pi : G_F \rightarrow GL_2(\bar{\mathbb{A}}_p) \text{ irreducible}$$

Fact. $V(\pi)^{\text{ss}} \simeq \otimes\text{-Ind } \rho_\pi$ tensor induction of ρ_π

Assume $F|A$ Galois, $\sigma \in \operatorname{Gal}(F|A)$ $\rightsquigarrow \rho_\pi^\sigma : G_F \rightarrow GL_2(\bar{\mathbb{A}}_p)$
 $\uparrow \quad \quad \uparrow$
 $\tilde{\sigma} \in G_A$ conjugate of ρ_π by $\tilde{\sigma}$

$$\otimes\text{-Ind } \rho_\pi = \bigotimes_{\sigma \in \operatorname{Gal}(F|A)} \rho_\pi^\sigma, \quad \dim = 2^d$$

Rem. If $\dim \rho_\pi = 1 \rightsquigarrow$ transfer in CFT.

Problem. $\otimes\text{-Ind } \rho_\pi$ reducible in general.

Th (Nekovář) $H_p^*(\mathbb{L}_k)$ is a s.s. G_A -rep.

Cor $V(\pi) = V(\pi)^{\text{ss}} \simeq \otimes\text{-Ind } \rho_\pi$

Main Th (Jiang-P.) $V(\pi) \simeq \otimes\text{-Ind } \rho_\pi$ if π is not CM

$$\begin{aligned} (\Rightarrow \rho_\pi &\neq \text{Ind } \chi \\ \Leftrightarrow \operatorname{Lie} \operatorname{Im}(\rho_\pi) &\supset \operatorname{sl}_2) \end{aligned}$$

Rem. We also have a version for completed cohomology.

Plectic structure

$$G_A := \text{Aut}(\bar{A} | A)$$

$$G_F^d \rtimes \mathcal{G}_d \simeq \overset{\text{plec}}{G_F} := \text{Aut}(\bar{A} \otimes_A F | F)$$

plectic Galois group

$$\bar{A} \otimes_A F \simeq \prod_{\tau: F \hookrightarrow \bar{A}} \bar{A}$$

$\cup$

$$G_F^d \rtimes \mathcal{G}_d$$

Conj. (Nekovář - Scholl) G_A -action on $H^*(\text{Sh}_K, \bar{A}, \mathbb{L}_K)$ can be ext.

to an action of G_F^{plec} .

$$P_\pi \xrightarrow{\text{plectic induction}} \underbrace{\text{Ind}^{\text{plec}} P_\pi}_{\text{irreducible}}: P_F^{\text{plec}} \rightarrow GL_d(\bar{\mathbb{Q}}_p) \text{ s.t. } \text{Ind}^{\text{plec}} P_\pi|_{G_A} = \otimes\text{-Ind } P_\pi$$

Nekovář: construct local plectic group action. partial Frobenius at $l \neq p$

Jiang - P.: construct "Lie G_F^{plec} " partial Sen operator.

Th. (Sen) $L \subset \bar{\mathbb{Q}}_p$, $L | \mathbb{Q}_p$ finite, $P: G_L \rightarrow GL_d(\mathbb{Q}_p)$, $\mathbb{C}_p = \widehat{\bar{\mathbb{Q}}_p}$

Sen constructed θ (Sen operator) $\in \text{gl}_d(\mathbb{C}_p)$

$$(1) \theta \in \text{Lie } P(I_L) \otimes \mathbb{C}_p$$

$$(2) \text{ If } V \subset \text{gl}_d(\mathbb{Q}_p) \text{ } \mathbb{Q}_p\text{-subspace s.t. } V \otimes \mathbb{C}_p \ni \theta, \text{ then } V \supset \text{Lie } P(I_L)$$

Construction of partial Sen operators

$$- F = \mathbb{Q}_1, \quad P.$$

$$- d > 1, \quad \text{Rodriguez Carmargo}$$

$$I_n = \int_0^1 x^n e^{-x} dx$$

$$I_n = \int_0^1 x^n e^{-x} dx = \left[-x^n e^{-x} \right]_0^1 + \int_0^1 n x^{n-1} e^{-x} dx$$

$$= -e^{-1} + n I_{n-1}$$

$$I_0 = \int_0^1 e^{-x} dx = \left[-e^{-x} \right]_0^1 = 1 - e^{-1}$$

$$I_1 = -e^{-1} + 1 I_0 = -e^{-1} + 1(1 - e^{-1}) = 1 - 2e^{-1}$$

$$I_2 = -e^{-1} + 2 I_1 = -e^{-1} + 2(1 - 2e^{-1}) = 2 - 5e^{-1}$$

$$I_3 = -e^{-1} + 3 I_2 = -e^{-1} + 3(2 - 5e^{-1}) = 6 - 16e^{-1}$$

$$I_4 = -e^{-1} + 4 I_3 = -e^{-1} + 4(6 - 16e^{-1}) = 24 - 65e^{-1}$$

$$I_5 = -e^{-1} + 5 I_4 = -e^{-1} + 5(24 - 65e^{-1}) = 120 - 325e^{-1}$$

$$I_6 = -e^{-1} + 6 I_5 = -e^{-1} + 6(120 - 325e^{-1}) = 720 - 1951e^{-1}$$

$$I_7 = -e^{-1} + 7 I_6 = -e^{-1} + 7(720 - 1951e^{-1}) = 5040 - 13657e^{-1}$$

$$I_8 = -e^{-1} + 8 I_7 = -e^{-1} + 8(5040 - 13657e^{-1}) = 40320 - 109255e^{-1}$$

$$I_9 = -e^{-1} + 9 I_8 = -e^{-1} + 9(40320 - 109255e^{-1}) = 362880 - 983295e^{-1}$$

$$I_{10} = -e^{-1} + 10 I_9 = -e^{-1} + 10(362880 - 983295e^{-1}) = 3628700 - 9732845e^{-1}$$

$$I_{11} = -e^{-1} + 11 I_{10} = -e^{-1} + 11(3628700 - 9732845e^{-1}) = 39925700 - 107061295e^{-1}$$

$$I_{12} = -e^{-1} + 12 I_{11} = -e^{-1} + 12(39925700 - 107061295e^{-1}) = 479108400 - 1284735540e^{-1}$$